88. Consider the following pairs:

Region often mentioned in the news		Country
1. Anatolia	-	Turkey
2. Amhara	-	Ethiopia
3. Cabo Delgado	-	Spain
4. Catalonia	-	Italy

How many pairs given above are correctly matched?

(a) Only one pair (b) Only two pairs
(c) Only three pairs (d) All four pairs

Ans. (b) : Anatolia – Turkey, Amhara – Ethiopia
Cabo delgado – Mozambique, Catalonia – Spain

89. With reference to Indian laws about wildlife protection, consider the following statements:

1. Wild animals are the sole property of the government.
2. When a wild animal is declared protected, such animal is entitled for equal protection whether it is found in protected areas or outside.
3. Apprehension of a protected wild animal becoming a danger to human life is sufficient ground for its capture or killing.

Which of the statements given above is/are correct?

(a) 1 and 2 (b) 2 only (c) 1 and 3 (d) 3 only

Ans. (a) : As per section 39 of Wild Life Protection Act 1972 wild animals are government property. However Section 40A provides immunity in certain cases. Thus statement 1 is correct.
When an animal is declared as a protected animal under Wildlife (Protection) Act (WPA), 1972 then it will enjoy the same protection either if it is present in a protected area or outside of a protected area. Hence, statement 2 is correct.
Only if the wild animal becomes a danger to human life or is diseased or disabled beyond recovery it can be allowed to be captured or killed by the competent authority, the Chief Wildlife Warden of the State. This provision is applicable to wild animals listed in Schedule I of the Wildlife (Protection) Act, 1972, which includes leopards. Mere apprehension or fear that a wild animal could endanger human life is not a ground for capture or killing. Hence statement 3 is incorrect. UPSC has considered option (b) as correct.

90. Certain species of which one of the following organisms are well known as cultivators of fungi?

(a) Ant (b) Cockroach (c) Crab (d) Spider

Ans. (a) : Species of Ant is well known as the cultivator of the fungi. Few other insects such as termites, beetles, and marsh periwinkles are also able to cultivate fungi.
Hence, option (a) is correct.

91. Consider the following pairs:

Site of Ashoka's major rock edicts		Location in the State of
1. Dhauli	—	Odisha
2. Erragudi	—	Andhra Pradesh
3. Jaugada	—	Madhya Pradesh
4. Kalsi	—	Karnataka

How many pairs given above are correctly matched?

(a) Only one pair (b) Only two pairs
(c) Only three pairs (d) All four pairs

Ans. (b) :

Site of Ashoka's major rock edicts		Location in the State of
1. Dhauli	—	Odisha
2. Erragudi	—	Andhra Pradesh
3. Jaugada	—	Odisha
4. Kalsi	—	Uttarakhand

92. Consider the following pairs:

King		Dynasty
1. Nannuka	—	Chandela
2. Jayashakti	—	Paramara
3. Nagabhata II	—	Gurjara-Pratihara
4. Bhoja	—	Rashtrakuta

How many pairs given above are correctly matched?

(a) Only one pair (b) Only two pairs
(c) Only three pairs (d) All four pairs

Ans. (b) : Nannuka was the founder of the Chandela dynasty of India. He ruled in the Jejakabhukti region. Jayashakti was a 9th-century ruler from the Chandela dynasty of Central India.
Nagabhata II was an Indian Emperor from Gurjara-Pratihara dynasty. He ascended the throne of GurjaraPratihara dynasty after his father Vatsraja.
Bhoja popularly known as Raja Bhoj Parmar (reigned c. 1010–1055 CE) was an Indian King from the Paramara dynasty.

93. Which one of the following statements about Sangam literature in ancient South India is correct?

(a) Sangam poems are devoid of any reference to material culture.
(b) The social classification of Varna was known to Sangam poets.
(c) Sangam poems have no reference to warrior ethic.
(d) Sangam literature refers to magical forces as irrational.

Ans. (b) : The Tamil society during the Sangam period was broadly divided into several groups. In the beginning of the Sangam Age, The Tamil society was not organized on the basis of the Vedic caste system, namely Brahmanas, Kshatriyas, Sudras. However, the earliest of the Sangam literature, Tolkappiyam refers to the four divisions, which were prevalent in the Sangam society namely, Anthanar, Arasar, Vaislyar and Vellalar. It may be said that this classification roughly corresponds to the Vedic Social division.

94. "Yogavasistha"was translated into Persian by Nizamuddin Panipati during the reign of :

(a) Akbar (b) Humayun (c) Shahjahan (d) Aurangzeb

Ans. (a) : YogaVasistha was translated multiple times into Persian language throughout the Mughal Dynasty, as commanded by Akbar, Jahangir, and Darah Shikuh. Nizam al-Din Panipati completed one of these translations, known as the Jug-Basisht, in the late sixteenth century AD.

95. The world's second tallest statue in sitting pose of Ramanuja was inaugurated by the Prime Minister of India at Hyderabad recently. Which one of the following statements correctly represents the teachings of Ramanuja?

(a) The best means of salvation was devotion.
(b) Vedas are eternal, self-existent and wholly authoritative.
(c) Logical arguments were essential means for the highest bliss.
(d) Salvation was to be obtained through meditation

Ans. (a) : Ramanuja was deeply influenced by the Alvars. He firmly believed that intense devotion to Vishnu was the best means to attain salvation. He propounded the doctrine of Vishishtadvaita.
The outdoor 216-ft Statue of Equality is the world's second tallest statue featuring a sitting posture. It is composed of 'panchaloha', a combination of five metals comprising gold, silver, copper, brass, and zinc. Born in 1017 in Sri Perumbudur, Tamil Nadu, Sri Ramanujacharya liberated millions from social, cultural, gender, educational and economic discrimination with the foundational conviction that every human is equal regardless of nationality, gender, race, caste, or creed.

96. The Prime Minister recently inaugurated the new Circuit House near Somnath Temple at Veraval. Which of the following statements are correct regarding Somnath Temple?

1. Somnath Temple is one of the Jyotirlinga shrines.
2. A description of Somnath Temple was given by Al-Biruni
3. Pran Pratishtha of Somnath Temple (Installation of the present day temple) was done by President S. Radhakrishnan.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (a) : Statement 3 is incorrect: Pran Pratishtha of Somnath Temple (installation of the present day temple) was done by Dr. Rajendra Prasad and not by S. Radhakrishnan. Statement 1 and 2 are correct. Hence the correct answer is option (a).

97. Which one of the following statements best describes the role of B cells and T cells in the human body?

(a) They protect the body from environmental allergens.
(b) They alleviate the body's pain and inflammation
(c) They act as immunosuppressants in the body.
(d) They protect the body from the diseases caused by pathogens.

Ans. (d) : The overall ability of the body to fight the disease causing organisms (pathogens), conferred by the immune system is called immunity. The primary and secondary immune responses are carried out with the help of two special types of lymphocytes present in our blood, i.e., B-lymphocytes (B-cells) and T-lymphocytes (T-cells). Thus, both B-cells and T-cells are responsible for acquired immunity. B-cells are responsible for antibody-mediated immunity response. T-cells are responsible for cell-mediated immunity response. Bcells produce antibodies to fight with the pathogens into our blood. The T-cells themselves do not secrete antibodies but help B cells produce them. Hence option (d) is the correct answer.

98. Consider the following statements:

1. Other than those made by humans, nanoparticles do not exist in nature.
2. Nanoparticles of some metallic oxides are used in the manufacture of some cosmetics.
3. Nanoparticles of some commercial products which enter the environment are unsafe for humans.

Which of the statements given above is/are correct?

(a) 1 only (b) 3 only (c) 1 and 2 (d) 2 and 3

Ans. (d) : Nanoparticles are incredibly tiny particles of a given substance. Nanoparticles are less than 100 nanometers thick. To give some perspective, a nanometer is 1000 times smaller than the thickness of one strand of hair.
Nanoparticles can be naturally created, for example, minuscule droplets of sea spray. However, most nanoparticles are created in the lab. For sunscreen, the nanoparticles are zinc oxide and titanium dioxide. These ingredients are broken down into ultrafine particles before being added to sunscreen. So, statement 1 is not correct and statement 2 is correct.
Inhaled particulate matter can be deposited throughout the human respiratory tract, and an important fraction of inhaled nanoparticles deposit in the lungs. Nanoparticles can potentially move from the lungs to other organs such as the brain, the liver, the spleen and possibly the foetus in pregnant women. So, statement 3 is correct.
Therefore, option (d) is the correct answer.

99. Consider the following statements:
DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

(a) 1 only (b) 3 only (c) 1 and 2 (d) 2 and 3

Ans. (d) : DNA barcoding deals primarily with species identification and has limited utility for lower taxonomic levels (e.g. subspecies, animal breeds, plant varieties).
DNA barcoding is a novel system designed to provide rapid, accurate, and automatable species identifications by using short, standardized gene regions as internal species tags. Hence, statement 2 is correct.
There is an increasing demand for the improvement of quality controls, hence addressing scientific research towards the development of reliable molecular tools for food analysis. DNA barcoding is a widely used molecularbased system, which can identify biological specimens, and is used for the identification of both raw materials and processed food. Hence, statement 3 is correct.

100. Consider the following:

1. Carbon monoxide 2. Nitrogen oxide
3. Ozone 4. Sulphur dioxide

Excess of which of the above in the environment is/are cause(s) of acid rain?

(a) 1, 2 and 3 (b) 2 and 4 only (c) 4 only (d) 1, 3 and 4

Ans. (b) : Acid rain refers to the ways in which acid from the atmosphere is deposited on the earth's surface. Oxides of nitrogen and sulphur which are acidic in nature can be blown by wind along with solid particles in the atmosphere and finally settle down either on the ground as dry deposition or in water, fog and snow as wet deposition.
Burning of fossil fuels (which contain sulphur and nitrogenous matter) such as coal and oil in power stations and furnaces or petrol and diesel in motor engines produce sulphur dioxide and nitrogen oxides. SO_2 and NO_2 after oxidation and reaction with water are major contributors to acid rain, because polluted air usually contains particulate matter that catalyze the oxidation. Therefore, option (b) is the correct answer.